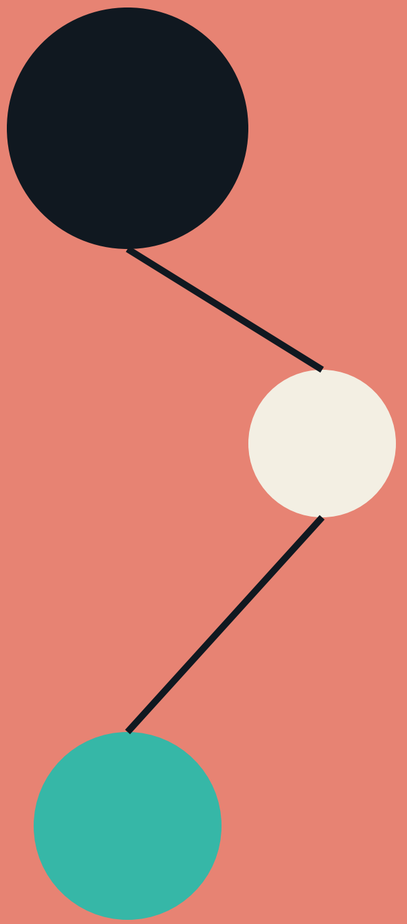
IS A SYSTEMATICALLY SCREENED INTERACTOME DIFFERENT FROM A CURATED ONE?

HuRI vs Lit-BM



Luck et al. 2020 · CB5313

Author · Course

Question and network construction

Does systematic screening recover a network with a different topology from literature curation?

SYSTEMATIC · HuRI



CURATED · Lit-BM

Published binary interactions → author and assay filters → protein identifiers → undirected edges

Ascertainment bias: studied proteins accumulate edges because they are studied—not necessarily because they are uniquely central.

Assumptions and limitations

BINARY ≠ CONTEXTUAL

An edge records a detectable pairwise interaction, not tissue, abundance or direction.

Y2H HAS A VIEWPOINT

Nuclear reporter, construct orientation and assay sensitivity shape detection.

ONE ISOFORM PER GENE

Isoform-specific interactions and proteoforms are collapsed.

COVERAGE IS INCOMPLETE

Absence of an edge is not evidence of biological non-interaction.

Computational workflow and research extension

A reproducible comparison with explicit sensitivity analyses.



RESEARCH EXTENSION

Density-match HuRI to Lit-BM, repeat random subsamples, and report uncertainty—not one convenient draw.

Global topology: two different network landscapes

Illustrative analysis snapshot; recompute from the submitted edge lists for final reporting.

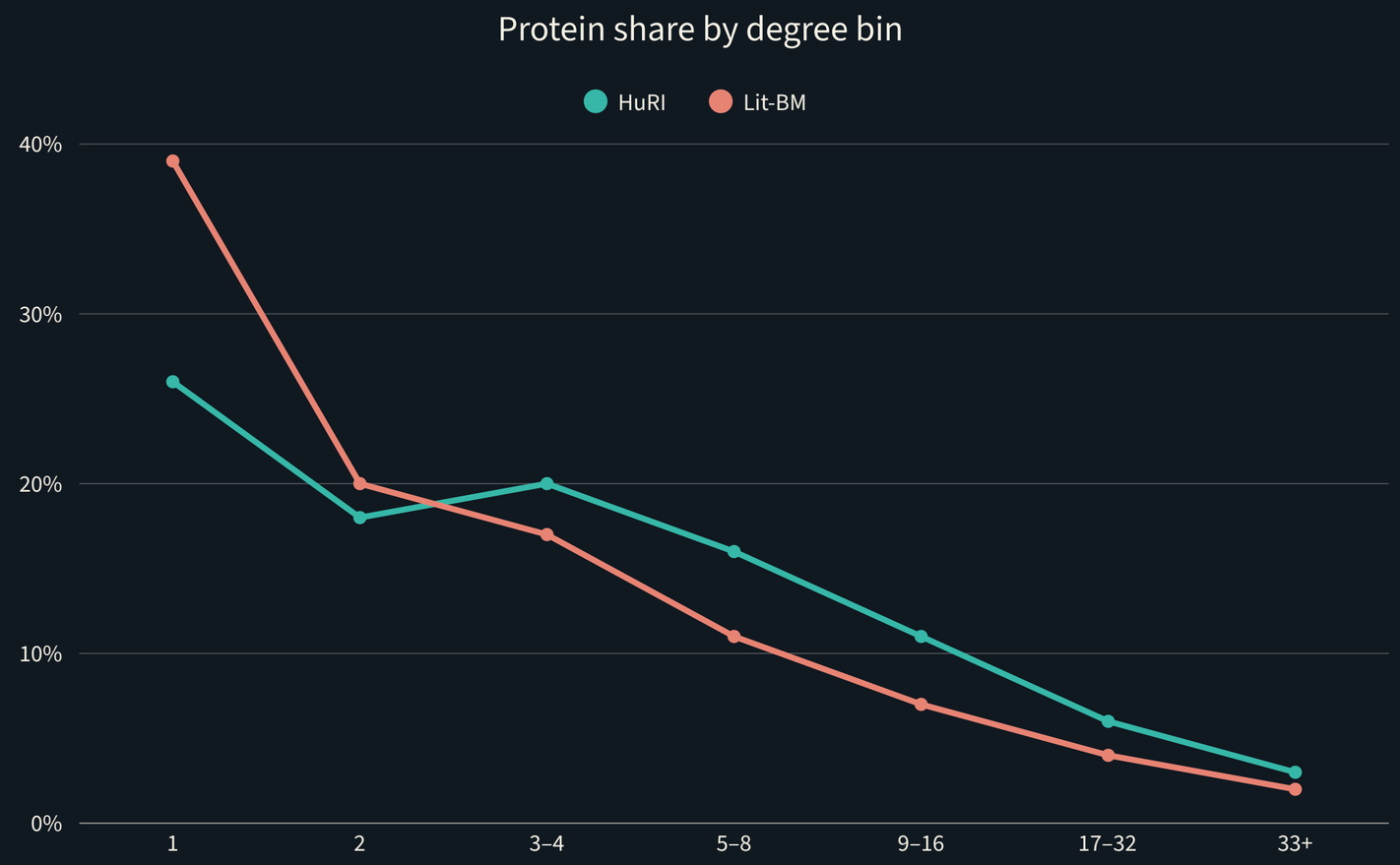
METRIC	HuRI	Lit-BM
Nodes	8,275	5,918
Edges	52,569	13,030
Mean degree	12.7	4.4
LCC	~99%	~91%
Path length	3.6	4.7
Clustering	0.047	0.091
Assortativity	−0.10	−0.17

More edges.
Less local closure.

Scale and sampling design change what ‘typical’ topology looks like.

Degree distribution and hub concentration

Both are heavy-tailed; literature curation concentrates attention more strongly.



DEGREE = 1

26% / 39%

TOP 1% EDGE SHARE

~16% / ~28%

LARGEST HUB

systematic / study-enriched

Clustering beyond degree: degree-preserving null

Thirty independent rewired graphs preserve every node’s degree while randomizing partners.

Clustering

0.047

0.031 ± .002

Assortativity

−0.10

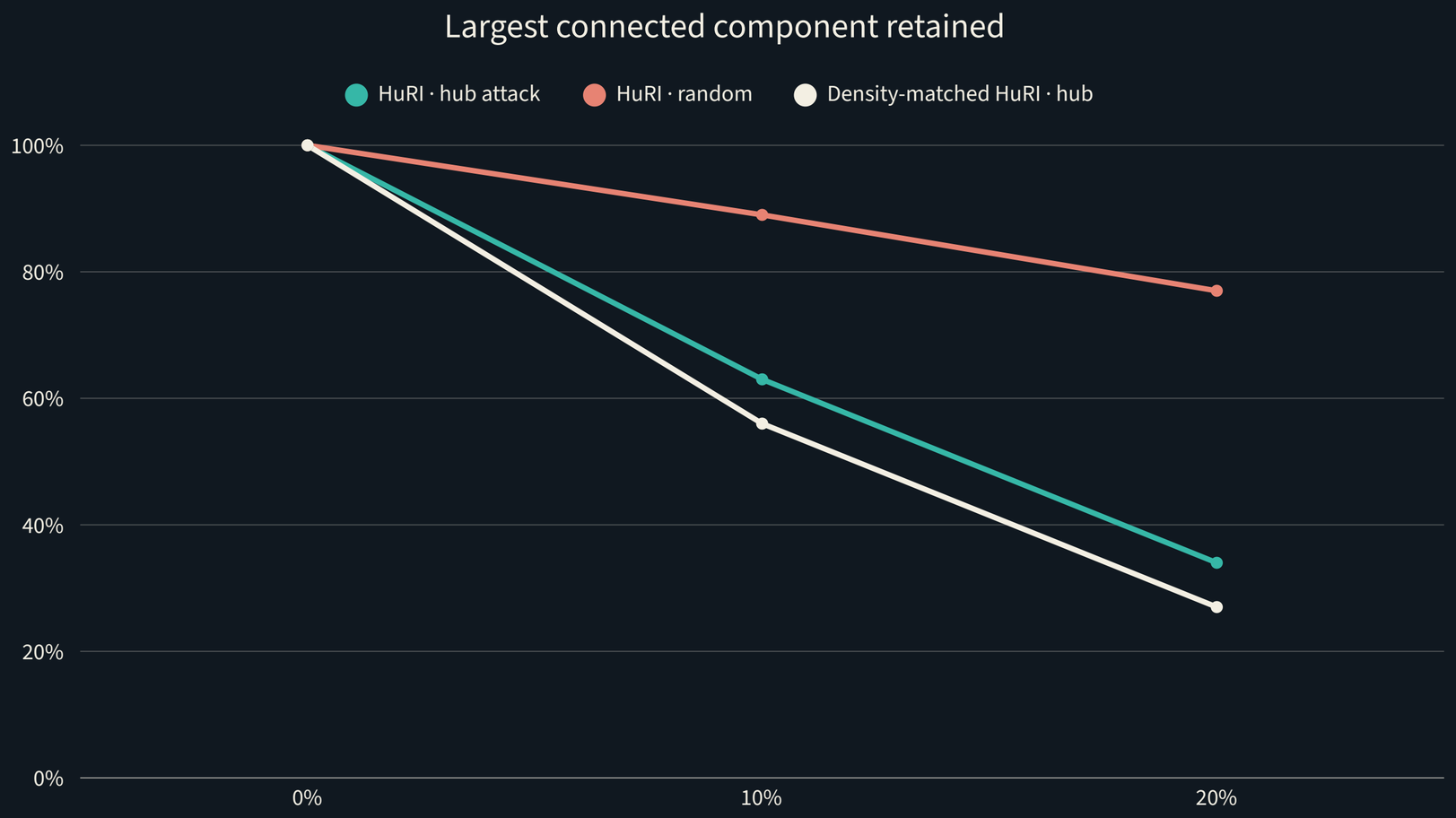
−0.08 ± .01

OBSERVED

Degree sequence is controlled. Residual clustering therefore reflects partner arrangement—not hub abundance alone.

Robustness: hub attack and density control

Largest-component retention after targeted versus random node removal.



Targeted removal exposes dependence on hubs.

Random removal tests redundancy.

Density matching separates sampling depth from architecture.

Core structure and shared-protein comparison

Compare architecture on the intersection—not only on each network’s full vertex set.

MAX k-CORE

HuRI 19

Lit-BM 11

SHARED PROTEINS

≈ 3.7k

common IDs

DEGREE CORRELATION

ρ ≈ .32

modest

TOP-HUB OVERLAP

low

sampling-specific

The same proteins can occupy different structural roles when the observation process changes.

Interpretation: what the analysis supports

01

Sampling strategy leaves a topological signature.

HuRI is larger and less locally clustered; Lit-BM is more concentrated around studied proteins.

02

Degree explains some—but not all—structure.

Degree-preserving nulls test whether closure and mixing exceed what hubs alone create.

03

Robustness conclusions require matched controls.

Density and coverage alter attack curves, so subsampling is part of the claim—not an appendix.

Direct test of study bias: model Lit-BM degree using publication count, gene age, essentiality and assay history.

THE NETWORK IS ALSO A RECORD OF HOW WE LOOKED.

Systematic screening and literature curation recover overlapping biology through different observation processes. Compare them with harmonized graphs, degree-aware nulls and density-matched controls.

REFERENCES

Luck K et al. A reference map of the human binary protein interactome. Nature (2020).

HuRI / CCSB Human Reference Interactome data portal.

Course analysis scripts and processed edge lists.

AI disclosure: generative AI assisted slide design and copy editing; scientific claims require author verification.